Interval-Membership Preservation under Bounded Score Perturbations

Research Architecture Runtime
`operator/interval-membership`

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Abstract

Closed interval indicator $I(x) = \mathbf{1}\{L \leq x \leq U\}$ under $|x' - x| \leq \varepsilon$. Lean `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

1 Theorems

Theorem 1 (Preservation). *If $L \leq U$ and $|x' - x| \leq \varepsilon$: (1) $L + \varepsilon \leq x \leq U - \varepsilon \Rightarrow I(x') = 1$; (2) $x < L - \varepsilon$ or $x > U + \varepsilon \Rightarrow I(x') = 0$.*

Theorem 2 (Sharpness). *If $L \leq x < L + \varepsilon$, then $x' = x - \varepsilon$ yields $I(x') = 0$.*

2 Comparison

Conjunction of two threshold sides; extends Threshold/Multi-threshold geometry to a band.

3 Lean / certificate

`Research.Operators.IntervalMembership.Preservation`; `lean/certificates/interval-membership/inter`

4 Limitations

Mathlib $\mathbb{R}$ (`REAL_MATHLIB`); axiom closure may include `Classical.choice` via `decide` on a conjunction.